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Numerical modeling of laminar flow over a porous cylinder under endothermic steam methane reforming reaction

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ABSTRACT

This paper deals with the numerical simulation of laminar flow over a porous cylinder under endothermic steam methane reforming reactions. The two-dimensional RANS approach is used to understand the effect of endothermic chemical reactions on the Kármán vortex street and heat transfer coefficient for a wide range of governing temperatures relevant to industrial applications of steam methane reforming. To achieve this goal, a set of calculations is performed for both transient and steady-state regimes, as well as for reactive and non-reactive flows. It was observed that steam methane reforming reactions have an effect on the Kármán vortex parameters. An increase in the catalytic cylinder temperature leads to an increase in the size of the single vortex and the length of the period. Under the analyzed conditions, for a cylinder temperature of 1200 K, the effect of chemical reactions on the Kármán vortex is maximal because the reaction rates strongly depend on temperature. Visualizations of the Kármán vortex formation for reactive and non-reactive flows are provided. Particular attention is paid to the analysis of the heat transfer coefficients on the cylinder surface. It was shown that endothermic chemical reactions significantly increase the heat supplied from the surface to the reacting flow.

1. Introduction

The numerical simulation of vortex dynamics in non-stationary laminar transverse flow around a circular cylinder, commonly referred to as the Kármán vortex street, has long captivated researchers due to its complex interplay of fluid mechanics and vortex shedding phenomena (Ledda et al., 2018; Amalia et al., 2018). This phenomenon not only serves as a fundamental example in fluid dynamics but also has practical implications across various engineering applications, including aerodynamics, hydrodynamics, and the design of structures subjected to fluid flow. The periodic formation and shedding of vortices can lead to oscillatory forces that affect the stability and performance of structures, making it crucial to understand these dynamics for predicting and mitigating potential issues (Dynnikova et al., 2021). Moreover, the rich variety of flow patterns generated by varying parameters such as Reynolds number and cylinder geometry provides a fertile ground for theoretical exploration and computational advancements, thus continually paying attention both academic researches and technological innovation.

The publications on the numerical simulation of the Kármán vortex street cover many practically interesting cases, varying Reynolds number, geometry, type of fluids, etc (Hammer et al., 2014; Kou et al., 2017; Demori et al., 2017). There are numerous comprehensive reviews on this topic (Zhang et al., 2018). Recently, the development of modern tools for numerical simulation has made it possible to study various aspects of transient laminar cross flow for a wide range of operational and design parameters (Martins et al., 2018; Ribeiro et al., 2020). Many papers address isothermal and non-isothermal flows over a cylinder (Jogee et al., 2020; Molla et al., 2006). Early research focused on the basic principles of heat transfer in laminar flow around cylinders, examining the influence of Reynolds numbers and Prandtl numbers on heat transfer rates. These studies often employed computational fluid dynamics (CFD) to simulate flow patterns and temperature distributions (Stringer et al., 2014; Ahn et al., 2017).

Recent papers have expanded on the transient aspects of heat transfer, highlighting the time-dependent behavior of thermal fields around a cylinder. Researchers have utilized unsteady-state simulations to capture the effects of varying inlet conditions and transient thermal loads (Yagmur et al., 2020; Kumar et al., 2015; Raza et al., 2021). Some studies have explored how changes in cylinder geometry (e.g., diameter, surface roughness) affect heat transfer characteristics. These

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investigations often include parametric studies to assess the impact of different shapes on fluid dynamics and thermal performance (Yuce and Kareem, 2016; Kumar et al., 2016).

Several papers discuss methods to enhance heat transfer, such as surface modifications or the introduction of secondary flows. Numerical simulations have been used to evaluate the effectiveness of these techniques under laminar flow condition (Tusar et al., 2019; Kumar, 2011). Advanced modeling approaches have integrated heat transfer with fluid flow dynamics to study phenomena like natural convection effects and phase change materials, providing insights into complex interactions in transient scenarios (Adhikari et al., 2020; Sahoo et al., 2017).

The Kármán vortex street is also an interesting process for flow with chemical reactions, such as combustion, because the cylinder or other shapes improve the mixing of combustion components. Smith et al. reported the results of LES analysis of premixed flow past a V-gutter flameholder (Smith et al., 2007). The authors performed three types of analyses: (1) nonreacting flow with nonvitiated air, (2) reacting flow with nonvitiated air at various equivalence ratios leading to blowout, and (3) reacting flow with vitiated air at various equivalence ratios leading to blowout. The results correlated well with experimental data.

Wu reported the results of a numerical study on hydrogen catalytic combustion over a circular bluff body (Wu et al., 2022). The author compared the flow characteristics and the differences in conversion efficiency between non-reactive and gas-phase reaction flows. The results show that the cylinder can also serve as a partial catalytic flame holder to ignite and stabilize the gas-phase reaction.

The chemically reacting flows with combustion over a cylinder differ from non-reacting flows mainly due to significant temperature variations, as the combustion processes produce a substantial amount of heat. At the same time, the stoichiometric mole number of the initial components is close to or equivalent to that of the reaction products.

Steam methane reforming is a widely used process to produce both hydrogen and synthesis gas, and it is an endothermic process that occurs over catalytic particles (Karpilov and Pashchenko, 2023; Ngo et al., 2019). The numerical simulation of steam methane reforming is of great interest to the research community due to the complexity of the process, which includes chemical reactions, heat and mass transfer inside and outside of catalytic particles, flow dynamics, etc (Behnam and Dixon, 2017; Lu et al., 2021; Pashchenko and Eremin, 2021). The main chemical reactions for steam methane reforming are provided below:

$$CH_4 + H_2O = CO + 3H_2 - 206.1\ \text{kJ/mol}. \tag{1}$$

$$CO + H_2O = CO_2 + H_2 + 41.1\ \text{kJ/mol}. \tag{2}$$

$$CH_4 + 2H_2O = CO_2 + 4H_2 - 165.0\ \text{kJ/mol}. \tag{3}$$

According to thermodynamic analysis, only reactions (1) and (3) have a notable effect on heat and mass balance (Dixon, 2017). Therefore, as seen from Eqs. (1)–(3), the stoichiometric mole number of the initial components is twice that of the reaction products. Moreover, the reaction rate and reaction enthalpy are strongly dependent on temperature. In addition, steam methane reforming occurs over catalysts; for example, Ni-based catalysts with porous Al_2O_3 support are widely used in large-scale processes.

In this paper, we numerically investigate the Kármán vortex street for transient laminar cross flow of a porous Ni-based catalytic cylinder with an endothermic steam methane reforming reaction. A comparison of Kármán vortices for reacting and non-reacting flows is reported for a wide range of temperatures applied in industrial-scale steam methane reforming process. The numerical model includes the following processes: fluid flow, heat and mass transfer in catalyst and flow zones, intra-catalytic diffusion, chemical reactions, etc. Moreover, the effect of the Kármán vortex street on heat and mass transfer is analyzed.

Table 1
The mesh independence study.

Case	1	2	3	4	5	6
Mesh	10 475	25 249	50 485	100 645	200 288	401 097
C_d	0.52	0.50	0.49	0.48	0.47	0.47

2. Methodology

In this paper, we present a CFD model for chemically reacting flow over a porous catalyst cylinder. The model encompasses several processes, including fluid flow, heat and mass transfer within the catalyst and the gaseous flow zone, chemical reactions, and intra-catalyst diffusion. Both transient and steady-state regimes were examined for reacting and non-reacting flows to investigate the effect of Kármán vortex formation on heat and mass transfer.

It is important to note that the interest in this problem primarily stems from the relative simplicity of the computational domain, which positions it alongside popular benchmark problems for developing methods to solve the Navier–Stokes equations. As a result, there is a wealth of literature detailing computational methods, approximation schemes, solver configurations for both commercial and open-source codes, mesh independence studies, and more.

In this section, we focus primarily on the processes involved in the CFD model, as other aspects of this example are addressed in tutorial materials, such as those found in the ANSYS Fluent Tutorial (Fluent et al., 2011).

2.1. Geometry and mesh

The computational domain and mesh structure are depicted in Fig. 1. In order to allow for a time-efficient computation, a 2D computational domain is chosen. In the reality, Karman vortex street is three-dimensional (Ausoni et al., 2007). However, there is sufficient information in literature that 2D approach provides a good correlation with experimental data and results obtained for 3D approach (Sakamoto et al., 1993; Bruno et al., 2010). Therefore, 2D computational domain provides sufficient accurate for CFD modeling of Karman vortex street. In the computational domain, a hot steam-methane mixture is entering at velocity inlet (Fig. 1a). A cylinder presents as a porous Ni-based catalyst with given temperature. The parameters of catalyst porous medium and temperature ranges for steam-methane mixture and catalyst are provided below. The flow channel size is 1.0 × 0.7 m. The diameter of catalytic cylinder is 0.04 m. The side walls are set as isothermal.

Mesh structure is shown in Fig. 1b. A fine spatial democratization of computational domain is essential for an accurate prediction of the transient laminar cross flow of a porous cylinder with and without endothermic steam methane reforming reaction. The boundary layers are added to get y^+ parameter that meets $y^+ < 10$:

$$\overline{y}^+ = MAX\left(y^+; y^+_{lim}\right) \tag{4}$$

where y^+_{lim} is maximum allowed value for each element of the mesh.

The number of mesh cells were determined based on mesh independence study for the case without chemical reaction (Table 1). Drag coefficient C_d was chosen as a control parameter. Because of mesh independence study for Karman vortex example is widely discussed in the literature, we omitted detailed its description (Ali et al., 2009; Khan et al., 2019). The number of cell is about 200,000.

2.2. Governing equations

The equations of mass, momentum and energy conservation coupled with chemical kinetic, diffusion and porous medium models were solved via Ansys Fluent.

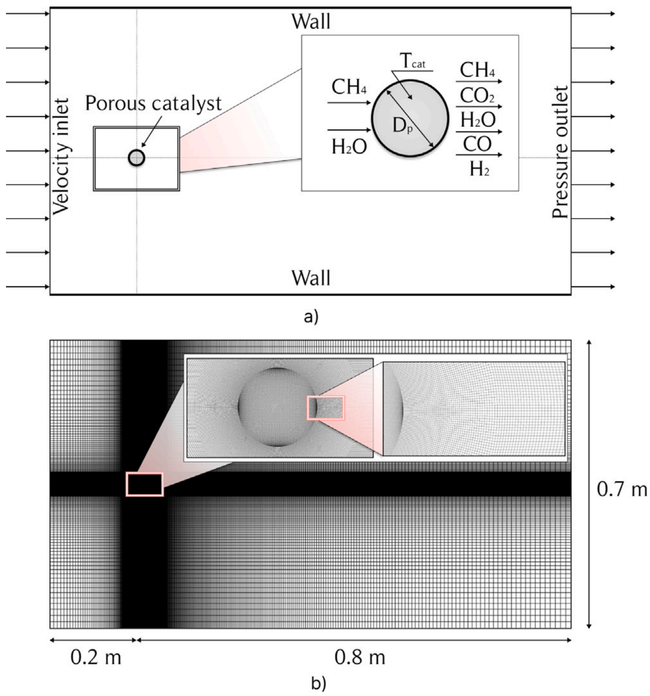

Fig. 1. Computational domain (a) and mesh (b).

The mass and momentum equations for transient flow (Batchelor, 1989):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \tag{5}$$

$$\frac{\partial}{\partial t}(\rho \vec{v}) + \nabla \cdot (\rho \vec{v}\vec{v}) = -\nabla \cdot p + \nabla \cdot (\overline{\overline{\tau}}) + \rho \vec{g} + \vec{F} + S_i \tag{6}$$

where p – the static pressure; $\rho\vec{g}$ – the gravitational body force; $\vec{F}$ – external body forces; S_i – momentum source term for porous medium.

Stress tensor $\overline{\overline{\tau}}$ in Eq. (6):

$$\overline{\overline{\tau}} = \mu \left[(\nabla \vec{v} + \vec{v}^{\mathrm{T}}) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \tag{7}$$

where μ – molecular viscosity; I – unit tensor, and the second term on the right hand side is the effect of volume dilation.

The energy equation is expressed as:

$$\frac{\partial}{\partial t}(\rho H) + \nabla \cdot \vec{u}(p + \rho \cdot h_f) = \nabla \cdot \left[k_{eff} \cdot \nabla T - \left(\sum h_j \cdot \vec{J}_j \right) + \overline{\overline{\tau}} \cdot \vec{v} \right] + S_f^h \tag{8}$$

where T – reaction mixture temperature; h_f – reaction mixture enthalpy; $\vec{J}_j$ – diffusion flux of j element; k_{eff} – conductivity; h_j – enthalpy of j element; S_f^h – gas enthalpy source term (include the source of energy due to chemical reaction).

The effective thermal conductivity of porous medium, k_{eff}, is computed as follows:

$$k_{\mathrm{eff}} = \epsilon_p k_f + (1 - \epsilon_p) k_s \tag{9}$$

where k_f, k_s – fluid phase and solid medium thermal conductivity, respectively; ϵ_p – porosity of catalytic particle.

The transition SST turbulence model is used to solve RANS (Menter, 1994).

A porous medium of catalyst particle is modeled by the addition of a momentum source term (S_i) to Eq. (6). The source term is composed of two parts: a viscous loss term and an inertial loss term. The actual pore size of catalyst particle is 10–100 nm. For such small pore size, the momentum that transported into particle is negligible. Therefore, the transport phenomena is defined by a permeability (α):

$$S_i = -\frac{\mu}{\alpha} v_i \tag{10}$$

Table 2
The coefficients for the equations of kinetic model.

Kinetics		
k_1	=	$5.922 \cdot 10^8 \exp\left(\frac{-209200}{R_g T}\right)$
k_2	=	$6.028 \cdot 10^{-4} \exp\left(\frac{-15400}{R_g T}\right)$
k_3	=	$1.093 \cdot 10^3 \exp\left(\frac{-109400}{R_g T}\right)$
Adsorption		
K_{H_2O}	=	$9.251 \cdot \exp\left(\frac{-15900}{R_g T}\right)$
K_{H_2}	=	$5.68 \cdot 10^{-10} \exp\left(\frac{93400}{R_g T}\right)$
K_{CO}	=	$5.127 \times 10^{-13} \exp\left(\frac{140000}{R_g T}\right)$
Equilibrium		
K_1	=	$1.198 \cdot 10^{17} \exp\left(\frac{-26830}{T}\right)$
K_2	=	$1.767 \cdot 10^{-2} \exp\left(\frac{4400}{T}\right)$
K_3	=	$2.117 \cdot 10^{15} \exp\left(\frac{-22430}{T}\right)$

A permeability (α) depends on pore size, tortuosity, etc. The permeability of Ni-Al_2O_3 catalyst is set as $2.8 \cdot 10^{-14}$ m^2 (Johnson et al., 1987). This catalyst consists of nickel oxide on a calcium aluminate support, typically containing 16% NiO, 1.5% SiO_3 and 0.05% SO_3. The catalyst properties are as follows: surface area of 14.3 m^2/g; skeletal density of 3190 kg/m^3; porosity of 0.44.

The chemical reactions of steam methane reforming (1)–(3) occur in porous catalyst and their reaction rates are determined based on Hou's kinetic model which is widely used for numerical simulation of steam methane reforming (Hou and Hughes, 2001):

$$R_1 = \frac{k_1 (p_{CH_4} p_{H_2O}^{0.5} - p_{H_2}^3 p_{CO} / K_1 p_{H_2O}^{0.5})}{p_{H_2}^{1.25} \cdot \mathrm{DEN}^2} \tag{11}$$

$$R_2 = \frac{k_2 (p_{CO} p_{H_2O}^{0.5} - p_{H_2} p_{CO_2} / K_2 p_{H_2O}^{0.5})}{p_{H2}^{0.5} \cdot \mathrm{DEN}^2} \tag{12}$$

$$R_3 = \frac{k_3 (p_{CH_4} p_{H_2O} - p_{H_2}{}^4 p_{CO_2} / K_3 p_{H_2O})}{p_{H_2}^{3.5} \cdot \mathrm{DEN}^2} \tag{13}$$

$$\mathrm{DEN} = 1 + K_{CO} \cdot p_{CO} + K_H \cdot p_{H_2}^{0.5} + K_{H_2O} \cdot p_{H_2O} / p_{H_2} \tag{14}$$

The expressions for coefficients in (11)–(13) are presented in Table 2:

The effective diffusion of ith species based on Bosanquet approximation:

$$D_i^{eff} = \left(\frac{1}{D_{m,i}^{eff}} + \frac{1}{D_{i,k}^{eff}} \right)^{-1} \tag{15}$$

where $D_{m,i}^{eff}$ – effective ordinary diffusivity; $D_{i,k}^{eff}$ – effective Knudsen diffusivity.

Effective ordinary diffusivity ($D_{m,i}^{eff}$) is defined from:

$$D_{m,i}^{eff} = \frac{\epsilon}{\tau} \frac{1 - x_i}{\sum \frac{x_i}{D_{ij}}} \tag{16}$$

where ϵ is porosity; τ is tortuosity; x_i is molar fraction of ith species; D_{ij} is binary diffusivity for i-jth pair.

Effective Knudsen diffusivity (D_{ij}) is calculated from:

$$D_{ij} = \frac{0.0143 T^{1.75}}{p_t M^{0.5} \left(V_i^{1/3} + V_j^{1/3} \right)^2} \tag{17}$$

where $V_{i,j}$ - diffusion volume for i, jth species (from (Fuller et al., 1966): H_2 = 7.07; H_2O = 12.7; CO_2 = 26; CO = 18.9; CH_4 = 16.8)

The effective Knudsen diffusivity is determined from:

$$D_{i,k}^{eff} = \frac{\epsilon}{\tau} 48.5 d_p \left(\frac{T}{M_i} \right)^{0.5} \tag{18}$$

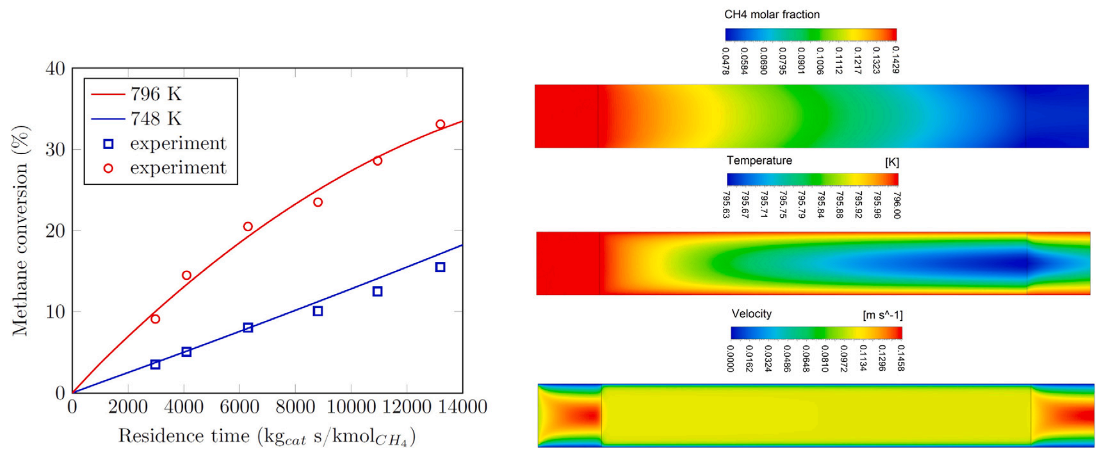

Fig. 2. (a) CH_4 conversion vs. residence time: solid lines — CFD modeling, hollow points — experimental results (Hou and Hughes, 2001); (b) Contours of CH_4 mole fraction, temperature and velocity at 796 K,p = 300 kPa, CH_4:H_2O:H_2 = 1:5:1, residence time 12000 $kg_{cat}s/kmol_{CH_4}$.

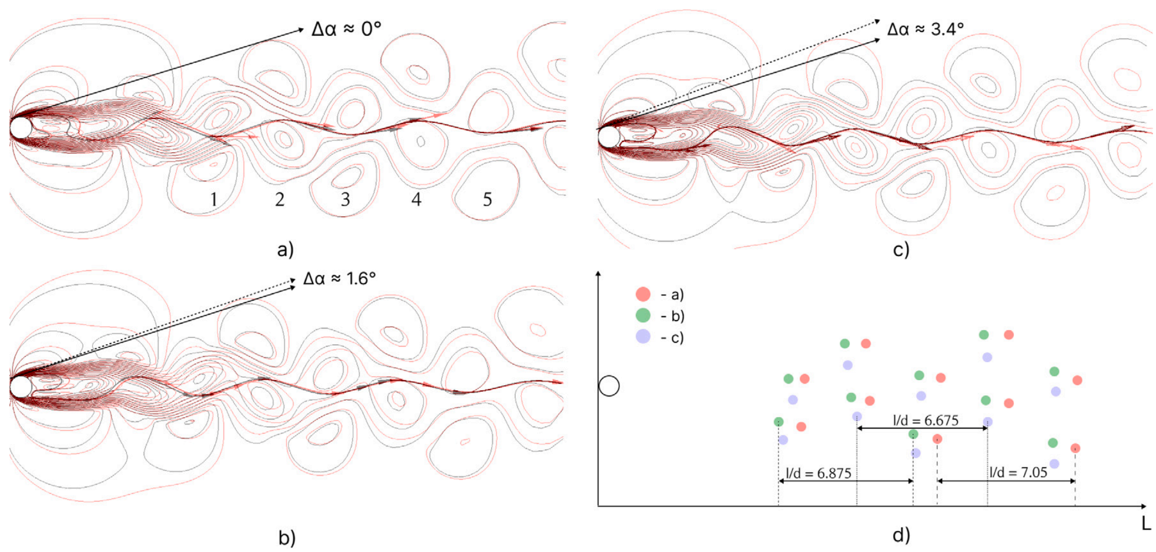

Fig. 3. Karman vortex street (contours of velocity) for T_{in} = 800 K when: (a) T_{cat} = 800 K; (b) T_{cat} = 1000 K; (c) T_{cat} = 1200 K; and position of vortex centers in the computational domain, (d). Black isolines — chemical reactions are off; red isolines — chemical reactions are on. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where d_p is mean pore diameter.

To implement chemical kinetic and diffusion model in Fluent solver, each model was converted into Used-Defined Function (UDF) written in C++ and uploaded to Fluent solver.

2.3. Solution procedure

The interaction of a uniform (with velocity u) flow of viscous incompressible fluid with a circular cylinder of diameter D_p is investigated for different Reynolds numbers. At the flow inlet, velocity inlet boundary conditions are set as follows: $u_x = u$, $v = 0$, $p_g = 0$. At the outlet, pressure outlet boundary conditions are chosen. The cylinder is set as isothermal heated to a given temperature. The temperature of inlet mixture varies for the given calculation tests.

The parameters of Fluent solvers are the following:

- Solver Type: Pressure-Based;
- Time: Transient;
- Operating Pressure: 101325 Pa;
- Solution Scheme: SIMPLE;
- Spatial Discretization:
  - Gradient-Gauss Node Based;
  - Pressure: standard;
  - Momentum, Energy, Turbulence, Species: Third-Order MUSCL;
- Controls for Species and Energy: 0.95;
- Residual:
  - Energy: 1e-06;
  - Others: 1e-05;
- Initialization: Standard;
- Time step size is 0.001 s (to get Courant number $\ll 1$)

To assess the effect of endothermic chemical reactions on Kármán vortex formation, a set of numerical tests was performed using the same model parameters, both with and without chemical reactions. To

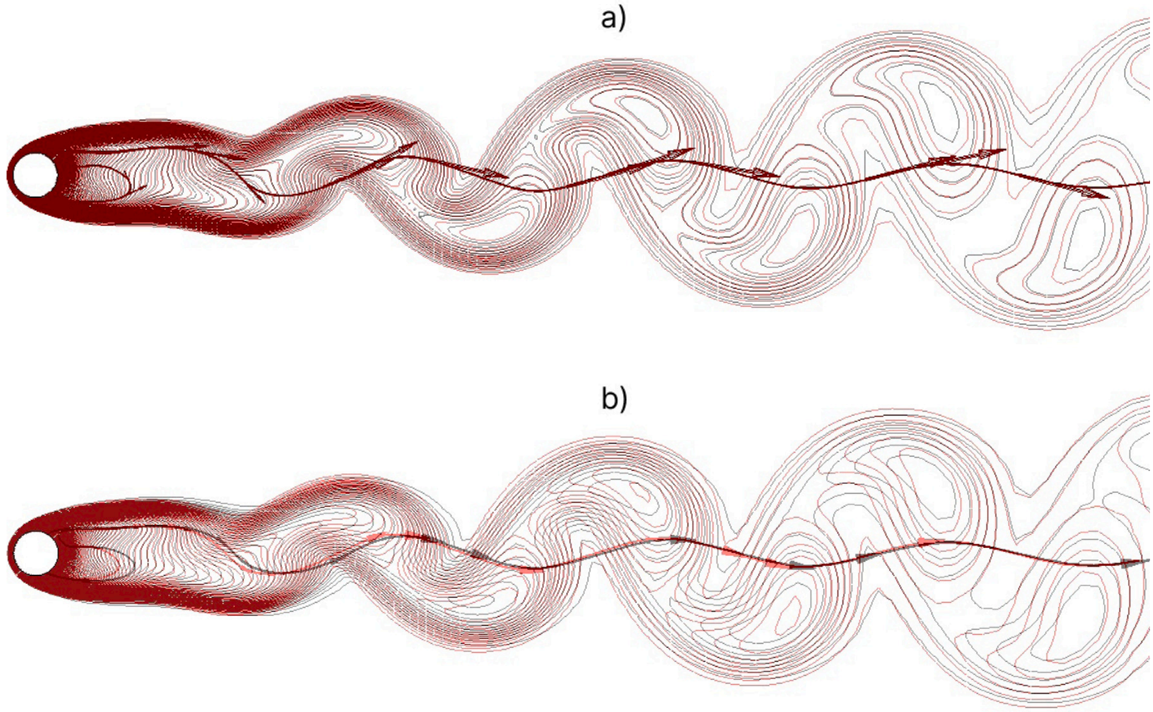

Fig. 4. Temperature isolines for Re = 1000, T_{in}=800 K when: (a) T_{cat} = 800 K; (b) T_{cat} = 1200 K.

understand the impact of Kármán vortex formation on heat transfer, another set of numerical tests was conducted for the same model parameters in both transient and steady-state regimes.

3. Results and discussion

3.1. Validation

The transient laminar cross flows over a cylinder were widely validated against experimental data and those results were presented in literature. However, there is no literature on the transient flows over a cylinder with endothermic chemical process of steam methane reforming. Therefore, in this section, we omitted validation of behavior of transient flow but focused on validation of heat and mass transfer for chemical process. To validate the parameters of heat and mass transfer for steam methane reforming over Ni-Al_2O_3 catalyst, the experimental results from Hou's paper were used (Hou and Hughes, 2001). The CFD model for validation was adopted for their experiments, where the reformer was filled with catalyst particles of 0.15–0.20 mm. The packed bed filled with the 0.15–0.20 mm spheres were simulated via pseudo-homogeneous model and source term (S_i) from Eq. (6) was calculated based on Ergun equation:

$$S_i = -\frac{\mu}{\alpha} v_i + C_2 \rho |v_i| v_i \tag{19}$$

$$\alpha = \frac{D_p^2}{150} \frac{\epsilon^3}{(1-\epsilon)^2}; \qquad C_2 = \frac{3.5}{D_p} \frac{(1-\epsilon)}{\epsilon^3} \tag{20}$$

where μ is kinematic viscosity; D_p is the mean particle diameter; ϵ is the packed bed porosity.

Fig. 2a shows a comparison of experimental data from Hou and Hughes' paper (Hou and Hughes, 2001) and calculated results for various temperaturesp = 300 kPa and CH_4:H_2O:H_2=1:5:1, when residence times varied by molar flow rates. Fig. 2a shows good correlation between the numerical and experimental results. The maximum deviation is 4.91% while the standard deviation is 3.86%.

Fig. 2b illustrates the distribution of methane molar fraction, temperature, and velocity at 796 K,p = 300 kPa, CH_4:H_2O:H_2 = 1:5:1, and a residence time of 12000 $kg_{cat}s/kmol_{CH_4}$ (to fully replicate the experiment from Hou's paper (Hou and Hughes, 2001)). The temperature variations do not significantly influence methane conversion because the temperature variations inside the fixed bed zone are less than 1 K, which is well correlated with the findings in Hou and Hughes' paper (Hou and Hughes, 2001). In that paper, the authors reported that the flow inside the fixed bed can be considered close to isothermal. Fig. 2 shows that the CFD model based on Hou and Hughes' kinetics can be used for numerical simulation of steam methane reforming over a catalytic cylinder to obtain accurate results.

3.2. Karman vortex street

The main goal of this paper is to understand the influence of the endothermic chemical process of steam methane reforming on Karman vortex street formation. It is well known that the steam methane reforming process produces a greater number of moles of synthesis gas than that of the initial products. For example, in the stoichiometric steam methane reforming reaction (1), the number of reaction products is twice that of the initial components. Additionally, steam methane reforming is endothermic, and heat consumption to compensate for this endothermic effect is required, which leads to changes in the heat and mass balance of the considered system.

Fig. 3 shows Karman vortex streets for various operational conditions at Re = 1000. The inlet gas temperature of 800 K is constant for all cases; however, the catalyst temperature varies: 800 K (a), 1000 K (b), and 1200 K(c). The simulation was conducted for two cases: chemical reactions are off (black isolines) and chemical reactions are on (red isolines). In addition, Fig. 3d depicts the positions of vortex centers and the dimensionless vortex rotation period. In Fig. 3d, l is the length that a vortex passes in one period, and d is the diameter of the cylinder.

Fig. 3 shows that accounting for chemical reactions affects the structure of the Karman vortex street. When the catalyst temperature

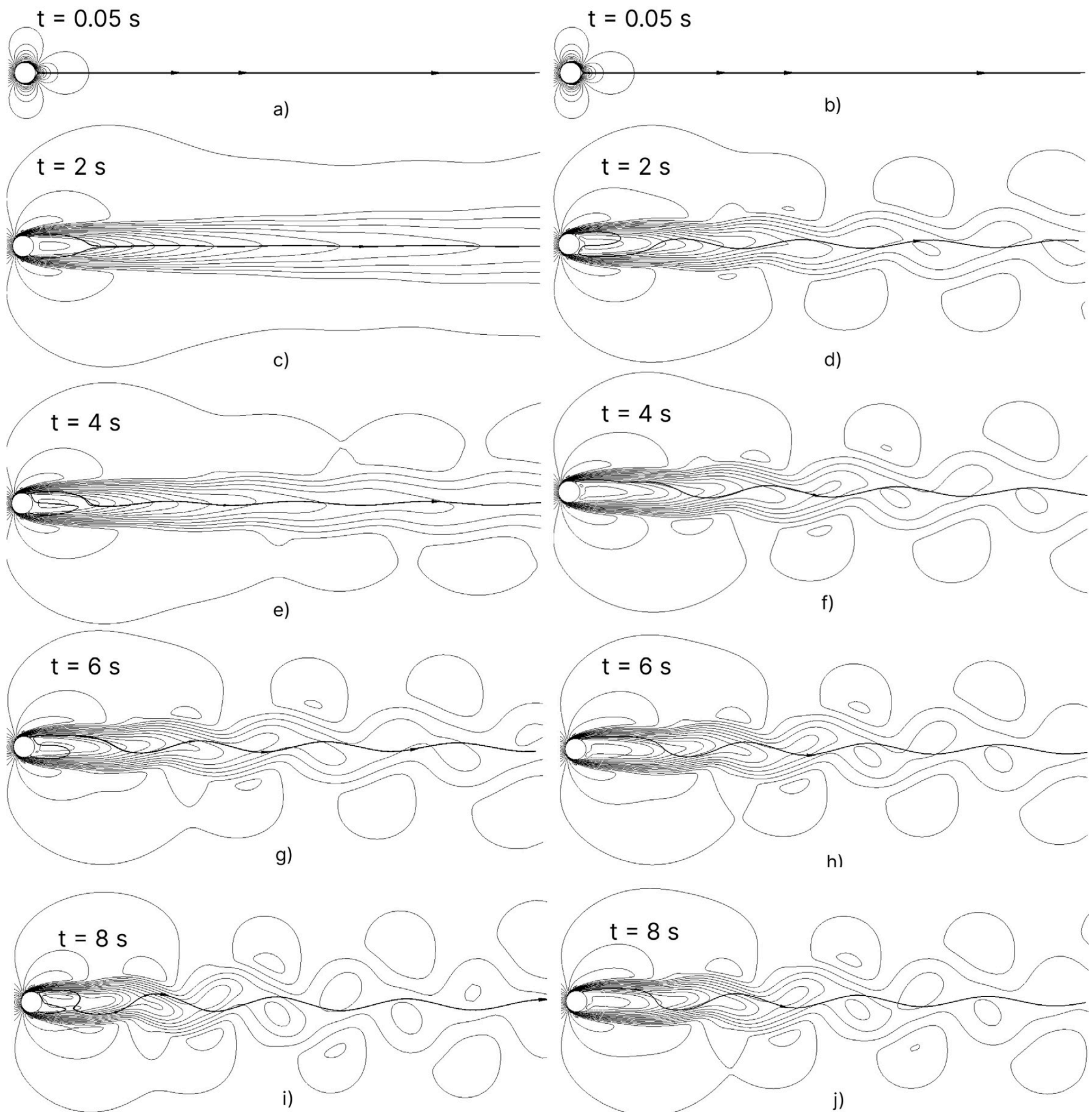

Fig. 5. Karman vortex street formation for Re = 1000, T_{in} = 800 K, T_{cat} = 1200 K: left side — chemical reactions are off; right side — chemical reactions are on.

is 800 K, this effect is minimal, and the isolines for both cases are similar. The main reason for this observation is that for T_{cat} = 800 K, the rates of reactions (1)–(3) are negligible, and the flow over the cylinder can be considered non-reactive. However, as the catalyst temperature increases, the effect of chemical reactions also increases. The maximal discrepancy is observed for T_{cat} = 1200 K because the increased temperature raises the reaction rates, resulting in a higher mole number of reaction products. Fig. 3d shows the quantitative effect of chemical reactions on the Karman vortex streets. An increase in catalyst temperature leads to an increase in the distance that a vortex travels in one period. The main reason for this observation is that higher reaction rate produces more reaction products. As a consequence, the number of moles of reaction products is higher than that of the initial components. As a result, the density of reaction products is lower than that of the initial components.

The chemical reactions also affect the character of the temperature isolines. Fig. 4 shows the temperature isolines for Re = 1000, T_{in} = 800 K when: (a) T_{cat} = 800 K; (b) T_{cat} = 1200 K. As can be seen from the figure, for T_{cat} = 800 K, the isolines for reactive and non-reactive flows are similar. This is due to the fact that the reaction rates are minimal, and both flows can be considered close to isothermal T_{cat} = 800 K; T_{cat} = 1200 K. However, when the catalyst temperature increases, the heat supplied from the catalyst is consumed by endothermic reactions, while for the non-reactive flow, heat flux is supplied to the gas flow. Therefore, we can conclude that taking into account the chemical reaction of steam methane reforming also affects the temperature distribution in the Karman vortex street.

Another interesting observation was made during the numerical simulations. When chemical reactions are active, the Karman vortex street develops more rapidly compared to when they are off. The velocity contours as a function of time for both non-reactive and reactive cases are shown in Fig. 5. As the catalyst temperature decreases, the differences between the Karman vortices for reactive and non-reactive flows become smaller. To further illustrate our findings, we have included a video displaying the velocity contours for both the reactive and non-reactive flows in Fig. 6 (multimedia available online).

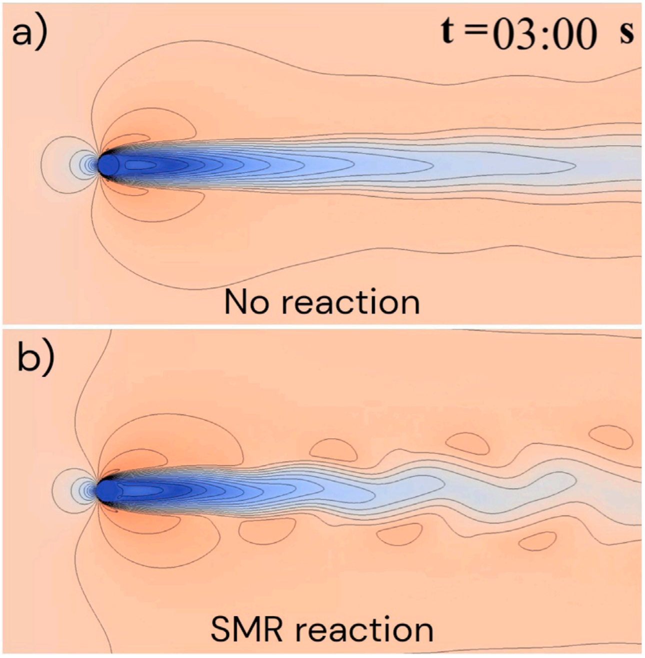

Fig. 6. Karman vortex street formation for Re = 1000, T_{in} = 800 K, T_{cat} = 1200 K: upper side — chemical reactions are off; down side — chemical reactions are on. Multimedia available online.

3.3. Heat transfer coefficient

In the previous subsection, we demonstrated the effect of endothermic chemical reactions on the Kármán vortex street. Obviously, the heat transfer between the catalytic cylinder and the steam-methane mixture is also influenced by chemical reactions. In this subsection, we discuss the impact of steam methane reforming reactions on the heat transfer coefficient. The steam methane reforming process was simulated over a wide range of temperatures. Moreover, we analyzed the heat transfer coefficient for both transient and steady-state flows as well as for reactive and non-reactive regimes.

To analyze the effect of chemical reactions of steam methane reforming on the heat transfer coefficient, a set of numerical calculations was conducted for reactive and non-reactive flows, as well as for transient and steady-state regimes. Fig. 7 shows the variation of the heat transfer coefficient as a function of cylinder temperature for various inlet gas mixture temperatures: T_{in} = 800 K; T_{in} = 700 K; T_{in} = 600 K. The results are presented for both reactive and non-reactive flow (solid and dashed lines, respectively) in transient (a) and steady-state regimes (b). The heat transfer coefficient was calculated from Newton's law, taking into account the endothermic effect of chemical reactions of steam methane reforming.

Fig. 7 indicates that the heat transfer coefficient for non-reactive flow is close to constant for all investigated cases. This is mainly due to the fact that the Reynolds number for all cases is the same, and coupled with minor temperature deviations, this leads to negligible variations in the Nusselt number. On the other hand, there is a significant discrepancy for reactive flow, primarily caused by the endothermic effects of the chemical reactions involved in steam methane reforming. An increase in temperature leads to an increase in the endothermic effect of the steam methane reforming process. Therefore, for the same Reynolds number, more heat is transferred from the catalytic cylinder to the reacting gas flow. Fig. 7 shows a non-linear dependence of the heat transfer coefficient on catalyst temperature because the reaction rate also non-linearly depends on temperature.

In addition, Fig. 7 depicts that a decrease in inlet gas mixture temperature leads to a decrease in the heat transfer coefficient. The main reason for this observation is the transport phenomena inside the catalyst cylinder. Lower gas temperatures result in lower reaction rates near the surface zone of the cylinder, while the transfer of reaction products from the center of the cylinder is limited by intra-catalyst diffusion limitations (Pashchenko, 2023).

Moreover, Fig. 7 illustrates the deviation between the heat transfer coefficients obtained for the same parameters in transient and steady-state regimes. We analyzed vortex formation behind the cylinder and observed more intensive vortex formation in the transient regime. Fig. 8 shows the velocity vectors for reactive (a, b) and non-reactive (c, d) flows at transient (a, c) and steady-state (b, d) regimes for Re = 1000; T_{in} = 800 K; T_{cat} = 1200 K.

As can be clearly seen from Fig. 8, the density of vectors near the cylinder zone is higher for the transient regime compared to steady-state. Therefore, it is expected that the heat transfer coefficient should be slightly higher for the transient regime rather than for steady-state. At the same time, when comparing velocity vectors for reactive and non-reactive flows, it can be seen that the density of vectors for reactive flow is higher than for non-reactive flow. The main reason for this is that chemical reactions produce products with lower density compared to the initial mixture (CH_4/H_2O vs H_2/CO). The result of such reactions is a higher velocity near the cylinder zone and as a consequence a higher heat transfer coefficient.

The results obtained in this paper can help researchers and engineers gain a deeper understanding of the effects of various variables on the performance of steam methane reforming. Future directions for this topic include analyzing the reforming performance of fixed-bed reactors under transient regimes based on the results from this study.

4. Conclusion

In this paper, numerical modeling of laminar flow over a porous cylinder undergoing endothermic steam methane reforming reactions at various temperatures was performed. The numerical model includes the following processes: fluid flow, heat and mass transfer in catalyst and flow zones, intra-catalytic diffusion, chemical reactions, porous medium model, etc. The two-dimensional RANS equations coupled with chemical kinetics were solved using Ansys Fluent. The chemical kinetics model was validated against experimental data, yielding an average error of 3.86%. It was established that the chemical reactions affect the structure of the Kármán vortex street. When the catalyst temperature is 800 K, this effect is minimal, and the velocity isolines for both reacting and non-reacting flows are similar. However, as the catalyst temperature increases, the effect of chemical reactions also intensifies. The maximum discrepancy is observed at T_{cat} = 1200 K because the increased temperature raises the reaction rates, resulting in a higher mole number of reaction products. Furthermore, it was noted that when chemical reactions are active, the Kármán vortex street develops more rapidly compared to when they are inactive. As the catalyst temperature decreases, the differences between the Kármán vortex streets for reactive and non-reactive flows diminish. In additions, the tests demonstrated that heat transfer between the catalytic cylinder and the reacting mixture is also influenced by chemical reactions. It was determined that the heat transfer coefficient for non-reactive flow remains nearly constant across all investigated cases, while for reactive flow, an increase in cylinder temperature leads to a nonlinear increase in the heat transfer coefficient. Additionally, the heat transfer coefficient was compared for transient and steady-state regimes.

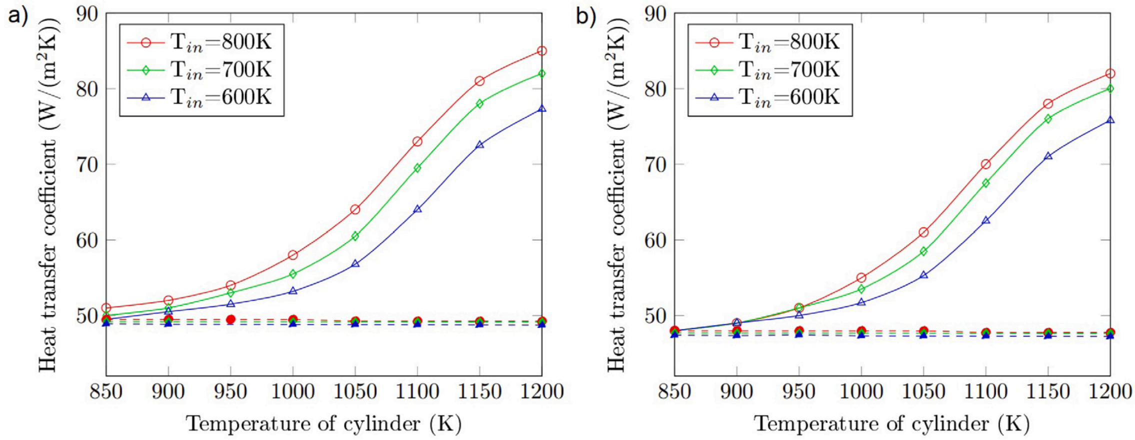

Fig. 7. The heat transfer coefficient as a function of cylinder temperature for various gas mixture temperature: solid lines — reactive flow; dashed lines — non-reactive flow; (a) transient regime; (b) steady-state regime.

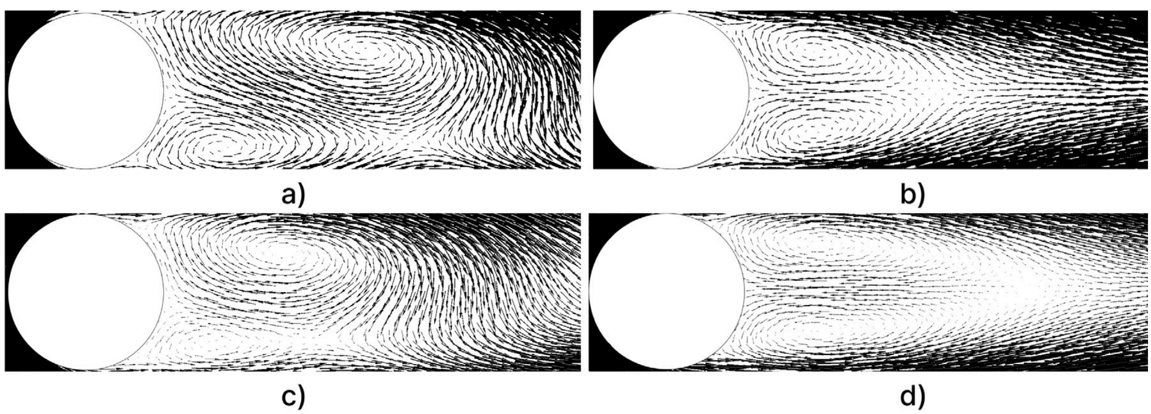

Fig. 8. Velocity vectors for reactive (a, b) and non-reactive (c, d) flows at transient (a, c) and steady-state (b, d) regimes for Re = 1000; T_{in} = 800 K; T_{cat} = 1200 K.

CRediT authorship contribution statement

Viacheslav Papkov: Visualization, Validation, Supervision, Software, Methodology, Investigation. **Boyan Zhang:** Validation, Software, Investigation. **Han Su:** Software, Investigation, Validation. **Haojie Chen:** Software, Investigation, Validation. **Dmitry Pashchenko:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijheatfluidflow.2024.109725.

Data availability

No data was used for the research described in the article.

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